

Amplitude modulation

Aim

1. To design and set up an amplitude modulator using transistor
2. To measure the modulation index for different amplitudes of the audio signal.

Components Required

Transistor - BC107

Resistors - $22k\Omega$, $1k\Omega$, $6.8k\Omega$, $10k\Omega$

Capacitor - $0.01\mu F$

Inductor - $100mH$

Theory

Amplitude modulation is obtained by varying the amplitude of the carrier by the modulating signal, the change in amplitude from the unmodulated value being proportional to the instantaneous value of the modulating signal independent of its frequency. Audio waves are not directly radiated from a transmitting station because it would require inconveniently large antennas. Also, large distances cannot be covered by audio frequencies. On the other hand, radio waves do not require very large antennas and they can be used over long distances. Hence the transmission of intelligence, viz. code, voice, music, etc., is accomplished by combining the electrical equivalent of the intelligence with a high-frequency wave at the transmitting end

The *modulation index*, *modulation factor*, or *depth of modulation*. It is defined as the ratio of the maximum deviation of the amplitude from the unmodulated value to the unmodulated carrier amplitude.

Thus the modulation index m_a can be determined from the modulated carrier waveform, by measuring V_{max} and V_{min} .

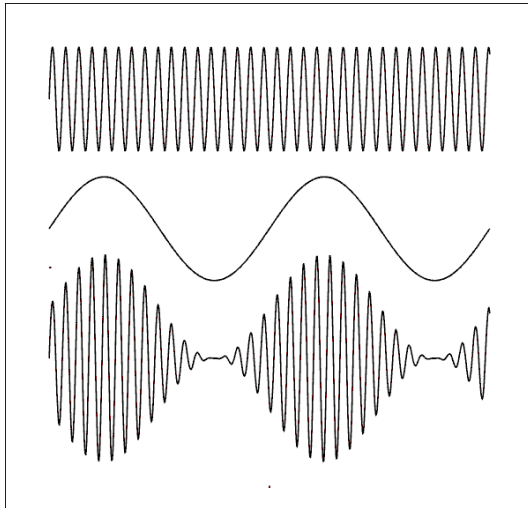
The quantity m_a (or M_a) gives the degree to which the signal modulates the carrier. The intelligence superimposed on the carrier is clear and stronger for a large value of m_a . Usually, m_a lies between 0 and 1. The maximum possible modulation without distortion is achieved for $m_a = 1$, known as 100 percent modulation. The amplitude of the modulated carrier is twice the unmodulated carrier amplitude on positive swings and zero on negative swings of the modulation envelope for 100 per cent modulation. *Overmodulation* occurs for $m_a > 1$. Overmodulation distorts the modulated signal and is therefore undesirable. The condition in which the peak values of the modulating signal stay for a long period of time well below the maximum value for which the transmission system or the equipment considered is designed, is known as under modulation.

$|m_a| < 1$ under modulation,

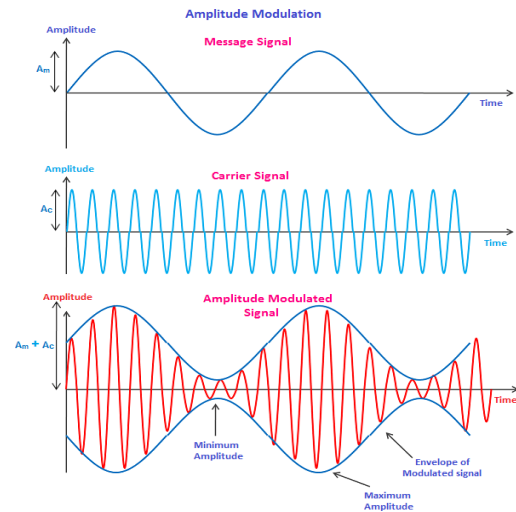
$|m_a| = 1$ critical modulation,

$|m_a| > 1$ over modulation.

Critical Modulation



Under Modulation



Over Modulation

